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practice-set -week-3

(Q1): λ: (I) $\Omega \in C$ (II) Closure under complementation (III) closed under finite & countable disjoint union

Take $A, B \subseteq \Omega$ s.t. $A \cap B \neq \phi$ (I) $A \not\subseteq B$ (II) $B \not\subseteq A$

$\{\phi, \Omega, A, B, A^c, B^c\} \rightarrow$ λ system but not σ-field

(Q2): Monotone class : countable increasing union (+) countable decreasing union

$\Omega = \{1,2,3\}$: monotone class: $\{\{1\}\}$.

(Q3)(a) $\mathcal{F}$: $\pi + \lambda$ system : π-system: $\mathcal{A}$: collection of subsets of Ω, Then $\mathcal{A}$ is closed under finite intersection

(I) Ω (II) point of λ system ✓ (III) To prove countable union property

MTH309 PRACTICE PROBLEMS WEEK 3

Question 1. Find an example of a λ-system which not a σ-field.

Question 2. Find an example of a Monotone class which not a σ-field.

Question 3. (a) If a collection $\mathcal{F}$ of subsets is both a π-system and a λ-system, then show that it is a σ-field.

(b) If a collection $\mathcal{F}$ of subsets is both a field and a Monotone class, then show that it is a σ-field.

Question 4. (Uniqueness of extension)

(a) Let $\mathcal{F}$ be a π-system on Ω. Suppose that $\mathbb{P}_1$ and $\mathbb{P}_2$ are two probability measures defined on the measurable space $(\Omega, \sigma(\mathcal{F}))$ such that they agree on $\mathcal{F}$, i.e.

$$\mathbb{P}_1(A) = \mathbb{P}_2(A), \forall A \in \mathcal{F}.$$

Show that the measures agree on $\sigma(\mathcal{F})$.

(b) Let $\mathcal{F}$ be a field on Ω. Suppose that $\mathbb{P}_1$ and $\mathbb{P}_2$ are two probability measures defined on the measurable space $(\Omega, \sigma(\mathcal{F}))$ such that they agree on $\mathcal{F}$, i.e.

$$\mathbb{P}_1(A) = \mathbb{P}_2(A), \forall A \in \mathcal{F}.$$

Show that the measures agree on $\sigma(\mathcal{F})$, i.e. $\mathbb{P}_1(A) = \mathbb{P}_2(A), \forall A \in \sigma(\mathcal{F})$.

Question 5. Let Ω be a countably infinite set, and let $\mathcal{F}$ be the field consisting of all finite subsets of Ω and their complements. Define

$$\mu(A) := \begin{cases} 0, & \text{if } A \text{ is finite} \\ 1, & \text{if } A^c \text{ is finite} \end{cases}.$$

Show that μ is finitely additive but not countably additive on $\mathcal{F}$, i.e. $\mathbb{P}_1(A) = \mathbb{P}_2(A), \forall A \in \sigma(\mathcal{F})$.

Question 6. (a) Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Show that f is continuous if and only if $f^{-1}(U)$ is open in $\mathbb{R}$ for all open sets U in $\mathbb{R}$.

(b) Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a function. Show that f is continuous if and only if $f^{-1}(U)$ is open in $\mathbb{R}^n$ for all open sets U in $\mathbb{R}^m$.

Question 7. For any subset A of Ω, define the indicator function of A to be $1_A : \Omega \to \mathbb{R}$ given by

$$1_A(\omega) := \begin{cases} 1, & \text{if } \omega \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Let $\{A_n\}_n$ be a sequence of subsets of Ω. For all $\omega \in \Omega$, prove the following statements.

$$\liminf_{n\to\infty} 1_{A_n}(\omega) = 1_{\liminf_{n\to\infty} A_n}(\omega),$$

and

$$\limsup_{n\to\infty} 1_{A_n}(\omega) = 1_{\limsup_{n\to\infty} A_n}(\omega).$$

Question 8. Let $E \in \mathbb{B}_{\mathbb{R}}$. Say that $A \subseteq E$ is open in E if there exists open $B \subseteq \mathbb{R}$ such that $A = B \cap E$. Show that

$$\mathbb{B}_E := \sigma(\text{all open sets in } E) = \mathbb{B}_{\mathbb{R}} \cap E.$$

Question 9. Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathbb{B}_{\mathbb{R}})$ be an RV. Consider the collection of sets

$$\sigma(X) := \{X^{-1}(A) : A \in \mathbb{B}_{\mathbb{R}}\}.$$

Check that it is a σ-field.

Question 10. Let $f : \Omega \to \Omega'$ be a function and let $\mathcal{F}$ be a collection of subsets of Ω'. Show that

$$\sigma(f^{-1}(\mathcal{F})) = f^{-1}(\sigma(\mathcal{F})),$$

where $f^{-1}(A)$ denotes the pre-image of $A \subset \Omega'$, $f^{-1}(\mathcal{F}) := \{f^{-1}(A) : A \in \mathcal{F}\}$ and $f^{-1}(\sigma(\mathcal{F})) := \{f^{-1}(A) : A \in \sigma(\mathcal{F})\}$. (Hint: Use the principle of good sets.)

$\{A_n\}_{n=1}^{\infty} \in \mathcal{F} \Rightarrow \bigcup A_n \in \mathcal{F}$

$\hookrightarrow B_1 := A_1$; $B_n := A_n \setminus \{A_1 \cup \dots \cup A_{n-1}\} \rightarrow \bigcup B_n = \bigcup A_n$

Today: B_i's are pw disjoint

$[B_i$'s $\in \mathcal{F}] \Rightarrow \bigcup B_n = \bigcup A_n$ (Proved) $\in \mathcal{F}$ (λ-system)

$B_i = A_i \cap \{A_1 \cup \dots \cup A_{n-1}\}^c$

$= A_i \cap A_1^c \cap \dots \cap A_{n-1}^c$ } finite complementation

By π-system: $B_i \in \mathcal{F}$

(b) Field + monotone class $\equiv$ σ-field

If $\{A_n\}_n$ be a sequence of subsets in $\mathcal{F}$.

Define: $\{B_n\}_n$ s.t. $B_1 := A_1$ & $B_n := A_n \setminus \{A_1 \cup \dots \cup A_{n-1}\}$

$C_n = \bigcup_{i=1}^{n} B_i$ Now, but $C_n \uparrow C = \bigcup_{i=1}^{\infty} B_i = \bigcup A_i$ (increasing)

If $A_i \in \mathcal{F}\ \forall i \Rightarrow B_i \in \mathcal{F}\ \forall i$ (By field def^n) $\Rightarrow C_i \in \mathcal{F}\ \forall i$

$\{C_n\}_n \in \mathcal{F}$ & by monotone class definition: $C \in \mathcal{F}$

$\bigcup B_i = \bigcup A_i$

Ans (4) (a): $\mathcal{F}$: πsystem on Ω

To prove $\mathbb{P}_1(A) = \mathbb{P}_2(A)$, $\forall A \in \mathcal{F}$

$\mathbb{P}_1(A) = \mathbb{P}_2(A)\ \forall A \in \sigma(\mathcal{F})$

* Approach: each $A \in \sigma(\mathcal{F})$ is a countable union/intersection/complement of $\mathcal{F}$. If we show that these conditions individually hold $\Rightarrow$ Proved. $\mathbb{P}_1(A) = \mathbb{P}_2(A) \Rightarrow 1 - \mathbb{P}_1(A) = \mathbb{P}_1(A^c) = \mathbb{P}_2(A^c)$ ✓

If $A_1, A_2 \in \mathcal{F} \Rightarrow A_1 \cap A_2 \in \mathcal{F}$ (π-system) $\Rightarrow \mathbb{P}_1(A_1 \cap A_2) = \mathbb{P}_2(A_1 \cap A_2)$ ✓

Similarly $\mathbb{P}_1(A_1 \cup A_2) = \mathbb{P}_2(A_1 \cup A_2)$ Follows.

(b) $\because \mathcal{F}$ is a σ-field $\Rightarrow$ closed under finite intersections $\Rightarrow$ π-system $\rightarrow$ (1) ✓

Ans (5): (I) Finite-Additivity

(i) If $\{A_i\}_{i=1}^{n}$ all are finite & disjoint $\Rightarrow \mu(\bigcup A_i) = 0 = \sum \mu(A_i)$

(ii) If atleast one say A_1 is infinite & all are disjoint

$\Rightarrow A_1^c$ is finite

$\Rightarrow$ For disjoint: $A_j \in A_1^c\ \forall j \geq 2 \Rightarrow$ only one can be infinite i.e. all others $\{A_i : i \geq 2\}$ is finite (else its complement would be infinite & mistake in disjoint)

Now: $\mu(\bigcup A_i) = 1 = \sum \mu(A_i)$

(II) Countably-additivity:

$\because \Omega$ is countably infinite $\Rightarrow$ enumeration $= \{\omega_1, \omega_2, \dots\}$

$\Rightarrow \mu(\Omega) = ??$ consider $\{A_n\}$ disjoint sets in $\mathcal{F}$ $\rightarrow$ countable

$\Rightarrow$ ~~A_i~~ $:= \{\omega_i\}$

Now $\mu(\Omega) = \mu(\bigcup A_i) \neq \sum \mu(A_i) = 0$ $\because A_i$ consists of only one element

$\because \Omega$ is finite)